H1 Master Skeleton Creation Report

17 September 2026 | Revision 01 | Dev-B: Layout / Architecture

Status: Initial master skeleton framework created and saved. The H1 headset now has a separate development model containing master layout references and tested associative cup-layout sketches. Full migration of the detailed components remains in progress.

MODEL RECORD	DETAIL
New Fusion model	H1 - Master Skeleton - DEV-B
Original assembly	H1 - Assembly - Updated - preserved; not overwritten
Master component	00 H1 MASTER SKELETON [DEV-B]
Material baseline	17-7 PH spring steel retained provisionally; future material change remains possible.



Figure 1. Saved development assembly. Master sketches are hidden in this view to show the preserved headset geometry; the image is not evidence of completed feature migration.

Purpose and business value

Establish shared design references so future architecture changes can propagate through dependent geometry. This first delivery provides the foundation for that workflow while retaining the existing headset as a reference. The regeneration test demonstrates the approach on a cup-layout sketch; it does not yet demonstrate automatic updates across the entire assembly.

Scope, verification and next steps

Completed in this delivery

ELEMENT	IMPLEMENTED SCOPE
Master parameters	22 user parameters covering envelope limits, cup layout, band targets, pivot-stop target and provisional cable allowances.
Layout geometry	Six overall envelope boundary planes, two cup layout datum planes, an XY envelope sketch and two cup envelope sketches.
Band and pivot references	A parameter-driven 73.8 mm band-radius target and two measured pivot-axis references. The existing multi-radius band was retained.
Associative references	Two separate cup-layout components with projections linked to master sketch geometry.
Separate model and exports	Saved under a new Fusion name, with full-assembly and skeleton-only local archive exports.

Verification results

CHECK	RECORDED RESULT
Parameter propagation	Cup A width: 102.252559 mm to 103.252559 mm. The downstream projected sketch followed the 1 mm change and was restored.
Assembly preservation	Original top-level occurrence bounding boxes matched within 0.001 mm after placement restoration. This is a placement/envelope check, not an exhaustive geometry comparison.
Timeline health	No feature warnings were reported in the final top-level timeline check.
File separation	The original and new Fusion documents have distinct document IDs. Both archive export calls completed successfully.

Remaining before the top-down conversion is complete

Connect existing detailed-part features to the master references and test each affected component. Resolve the exact sealing/interface planes, shell split lines, cable route, PCB envelopes and driver mounts. Confirm wearer left/right labels (currently Cup A/B), band-radius reference and pivot-stop interpretation. The 73.8 mm radius, 9.5 mm width midpoint, 17.5-degree stop midpoint and cable allowances must not be treated as approved manufacturing dimensions.

Delivery and review status

This report records creation of the **initial Dev-B framework**; it is not a completed architecture freeze or manufacturing release. The assembly retains 44 linked occurrences and its archive has not been verified as self-contained. A separate skeleton-only archive is available. GitHub publication was pending at report preparation because repository write permissions blocked the upload.

Evidence and references

Fusion API audit, save and projection-test records dated 17 September 2026; H1-DEV-B-Skeleton-Status.md; H1-Master-Skeleton-DEV-B.f3d; H1-Master-Skeleton-Only-DEV-B.f3d. Requirements basis: How We Build CAD at Aerosense (CAD Development Stages) and DEV H1 2026-08-17.md, with Stefan confirming the provisional spring-steel baseline. Prepared with Codex assistance for internal business review; no engineering approval is implied.